



User Manual

ZenMux-powered local AI office workspace



关于本手册

About this manual

This manual is based on the current **v0.6.86** source tree and real runtime UI. It covers installation, ZenMux configuration, Word, Excel, PowerPoint, Markdown, PDF, workbook SQL databases, cross-platform printing, and cross-editor AI workflows. Screenshots were captured on 16-17 August 2026 in an isolated temporary profile and contain no user documents, real API Key, or personal data. The research draft and workbooks are purpose-built synthetic demonstrations, not real scientific or business findings.

AI 适配与修改 / AI adaptation and modifications

AI adaptation and modifications by Zhiping Xu, College of Computer Science and Artificial Intelligence, Fudan University.

ZenOffice is derived from [genspark-ai/genoffice](#). It retains the Apache-2.0 foundation and adds ZenMux routing, local credential storage, academic workflows, cross-editor connections, and multi-platform distribution. Source and releases are available at [bennix/GenZenMuxAIOffice](#).

注意 • Notice AI 功能依赖网络、代理状态和模型服务可用性。AI 输出、联网结果、公式识别、审稿意见与图像生成均应由用户核对。AI features depend on network, proxy, and provider availability; always verify generated content, sources, formulas, reviews, and images.

Find the right section in one minute

Jump straight to the job you need to finish

This is the continuous English edition. Configure ZenMux, complete one three-minute recipe, then jump directly to Word, Excel, PowerPoint, Markdown, or PDF. Screenshots use the Chinese UI while paths name the matching controls.

Goal	Start here
Try it now	Three-minute quick wins after Part I
Use AI	Part II: selections, attachments, copy, and @Connect
Work with files	Parts III–VI: Word, Excel, PPT, Markdown, PDF
Research workflow	Equations, literature, citations, review, web verification
Fix a problem	Part VII and the one-minute troubleshooting table



- “Settings > AI models” means open Settings, then choose AI models.
- **Cmd/Ctrl** means Command on macOS and Ctrl on Windows/Linux.
- AI workflows require your ZenMux API Key; local-only editing does not.
- PDF focuses on viewing, pages, watermarks, security, and AI Q&A; its current ribbon does not expose text, image, or equation editing.

提示 • Tip Configure ZenMux first, then complete any three-minute recipe for your first end-to-end result.

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PART I

Install & configure

Start from a trusted build, complete first launch, and store the ZenMux API Key securely on this device.

Install and verify

安装并验证应用

ZenOffice ships as a macOS DMG, Windows x64 installer, Ubuntu DEB, and RPM. Download only from the project Landing Page or GitHub Releases.

1. Open the [official Landing Page](#), choose your platform, and confirm that the displayed version matches the filename.
2. macOS: open the DMG and drag ZenOffice to Applications. The Apple Silicon build is Developer ID signed; consult the matching Release verification notes for notarization-ticket status.
3. Windows: run the x64 installer. The current build is not Authenticode-signed, so SmartScreen may report an unknown publisher; verify the domain and Release asset first.
4. Ubuntu: run `sudo apt install ./zenoffice_0.6.86_amd64.deb`. Fedora/RHEL-family users can run `sudo dnf install ./zenoffice-0.6.86.x86_64.rpm`.

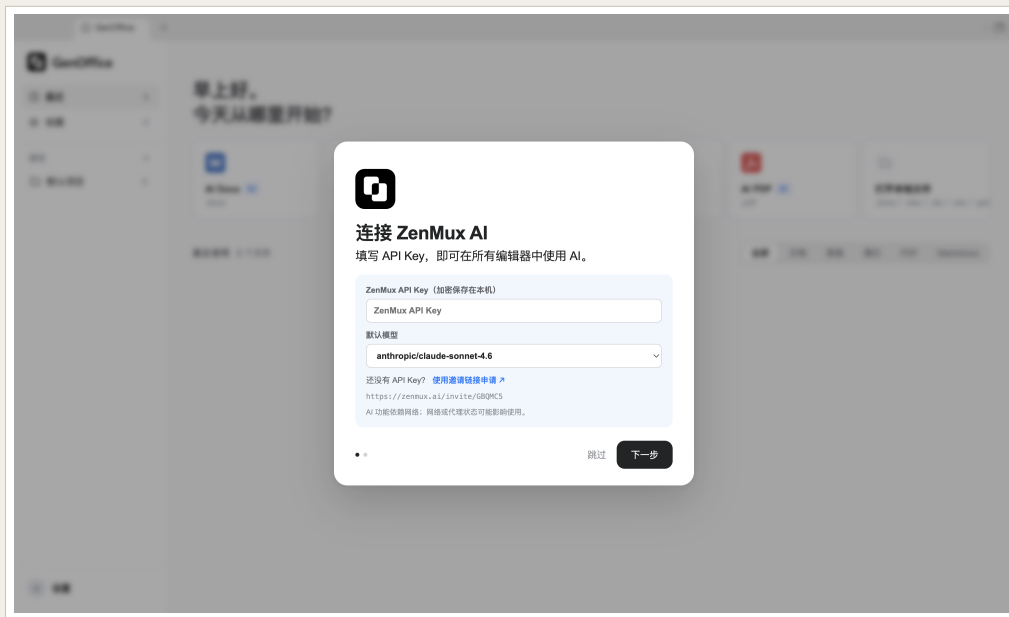
Platform	File	Requirement
macOS	.dmg	Apple Silicon, macOS 11+
Windows	.exe	Windows 10/11 x64
Ubuntu	.deb	Ubuntu 22.04/24.04 amd64
RPM	.rpm	Fedora/RHEL/Rocky/Alma/openSUSE x86_64

提示 • Tip The installer version, Landing Page label, and download target should agree. If a direct link fails, open GitHub Releases and choose the matching asset manually.

First launch and welcome

首次启动与欢迎页

The welcome screen connects directly to ZenMux. Enter an API Key now or skip and configure it later in Settings. Local file editing does not require sign-in.



Real runtime UI from the current source: ZenMux welcome screen with no real credential entered.

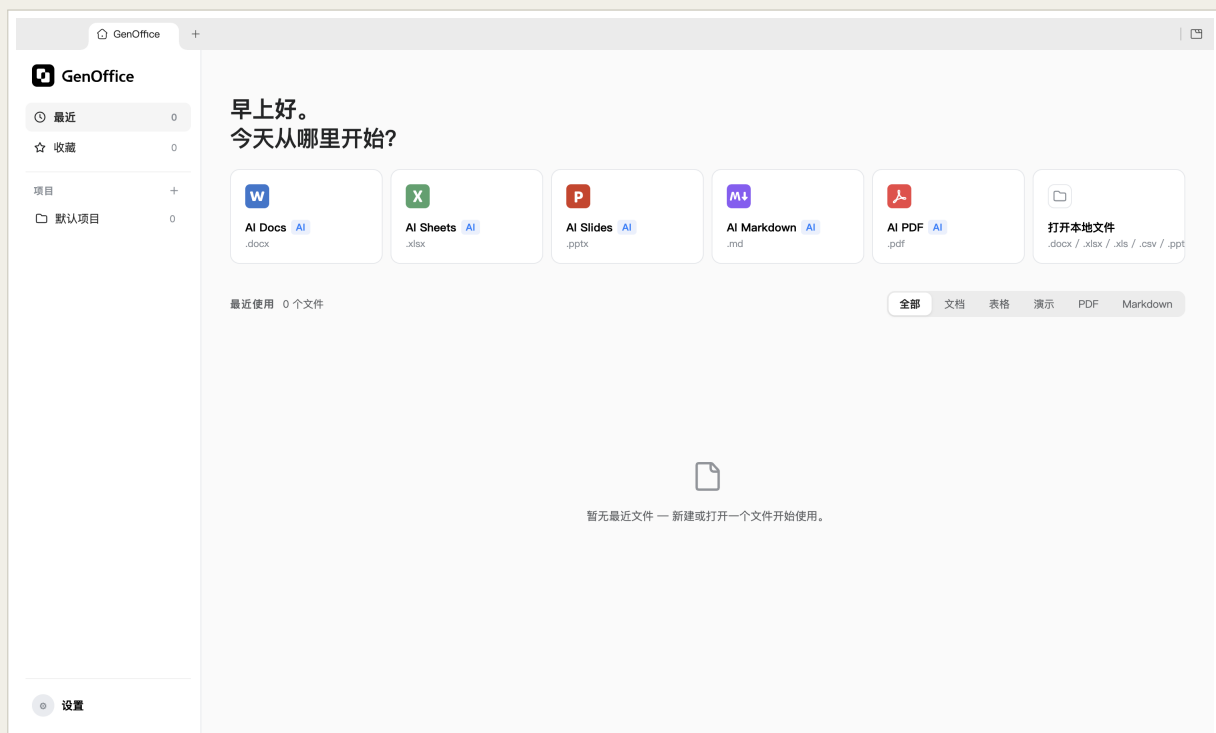
1. If you do not have a key, use <https://zenmux.ai/invite/GBQMC5> to register or request one.
2. Paste the key, choose the default text model, and continue. The value is masked in the interface.
3. To work locally first, choose Skip. Non-AI features in Word, Excel, PowerPoint, Markdown, and PDF remain available.

注意 • Notice Never place an API Key in document text, screenshots, chat attachments, or a public repository. Settings is the recommended entry point.

Home, projects, and recent files

主页、项目与最近文件

The home screen brings together five editors, local-file opening, projects, favorites, and recents. Tabs let one window host several file types.



Real runtime UI: quick-create cards, project navigation, recent-file filters, and Settings.

- Choose AI Docs, AI Sheets, AI Slides, AI Markdown, or AI PDF to create content; Open local file selects an existing file.
- You may also drop DOC/DOCX, XLS/XLSX/CSV, PPTX, PDF, or Markdown onto any module window. The desktop shell detects the actual format and routes it to the matching editor automatically.
- Projects organize work. Recent and Favorites provide quick return paths.
- The filter strip narrows recents to documents, spreadsheets, presentations, PDF, or Markdown.

提示 • Tip Dropping opens the file; it does not insert a copy into the current document. If a file is already open, switch to that tab instead of creating a second editing session.

Get AI ready: key, model, and connection test

快速配置 AI 算力与模型

Every AI path uses the OpenAI-compatible endpoint <https://zenmux.ai/api/v1>. Enter the key, choose a model, and run one short test to get started.



Real runtime UI: Settings > AI models, with an empty API Key field and no secret exposed.

1. Select Settings at the lower left of Home, then open AI models.
2. Paste the ZenMux API Key. The Base URL is fixed to <https://zenmux.ai/api/v1> and normally needs no change.
3. Choose one text model and one illustration model, then close Settings. Word, Excel, PowerPoint, Markdown, and PDF share the configuration.
4. Send a short, non-sensitive test question. If it fails, check network, proxy, quota, and model availability.

提示 • Tip The UI masks the key and saves it through operating-system secure storage. Enter a replacement in the same field to rotate credentials.

Add, select, and remove models

增加、选择与删除模型

Settings offers built-in models and accepts additional names in ZenMux provider/model-name form. Text and illustration models can both be removed locally.



Real runtime model area: current model, add-model field, illustration model, and delete controls.

- Built-in text models in this release: `anthropic/claude-sonnet-4.6`, `z-ai/glm-5v-turbo`, and `openai/gpt-5.4`.
- Formula image recognition model: `z-ai/glm-5v-turbo`.
- Illustration model: `openai/gpt-image-2`.
- Add the complete name shown by ZenMux, including the provider prefix, then select it from the current-model list.
- Delete removes only the local list entry; it does not delete a ZenMux model or account data. Keep at least one working text model.

提示 • Tip Image models may require different request formats. Prefer models already adapted by the app; if a new model rejects parameters, switch back to a built-in model to isolate the issue.

Verify privacy, version, and connectivity

确认隐私保护、版本与网络状态

The API Key is masked and protected by the operating-system credential service. Settings > About confirms version, adaptation credit, and update channel.



Real Settings > About runtime; the capture predates this release, so use the v0.6.86 header as the authoritative manual version.

- macOS uses operating-system secure storage/Keychain capabilities; Windows uses system credential encryption. The key is not embedded in installers or this manual.
- Opening, editing, and saving files are local-first. ZenMux calls, web search, literature search, formula OCR, and image generation require a network.
- Network, proxy, provider load, quota, or model status can affect speed and output. A client timeout does not prove that the provider produced no result.
- Use About to confirm the installed version. A useful issue report includes version, platform, action path, and a screenshot without secrets.

注意 • Notice If a key may be exposed, revoke or rotate it at ZenMux and replace it in Settings. Never send the key to support staff.

Three-minute quick wins

Skip the theory at first: choose one real task and produce an editable result in three minutes.

Recipe A: turn a screenshot into an editable Word equation

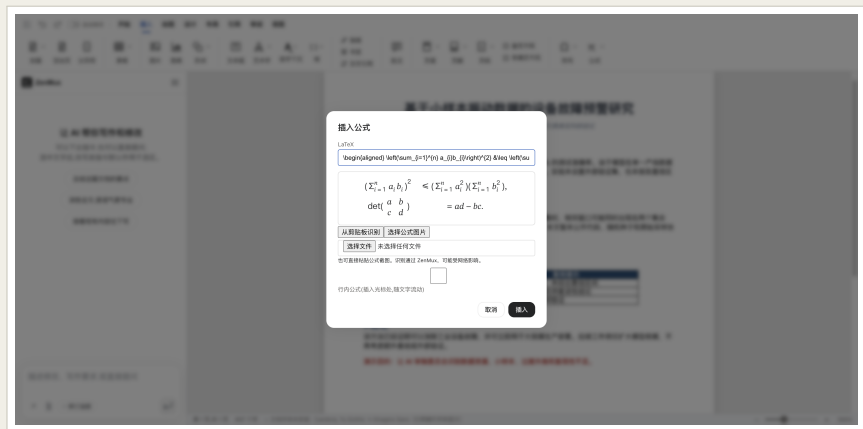
场景 A: 公式截图变成 Word 可编辑公式

Send a paper screenshot to ZenMux vision, verify the LaTeX, then insert an equation object that remains editable in Word.

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Synthetic input: Cauchy inequality and a 2×2 determinant



Real runtime result: editable LaTeX and rendered preview together

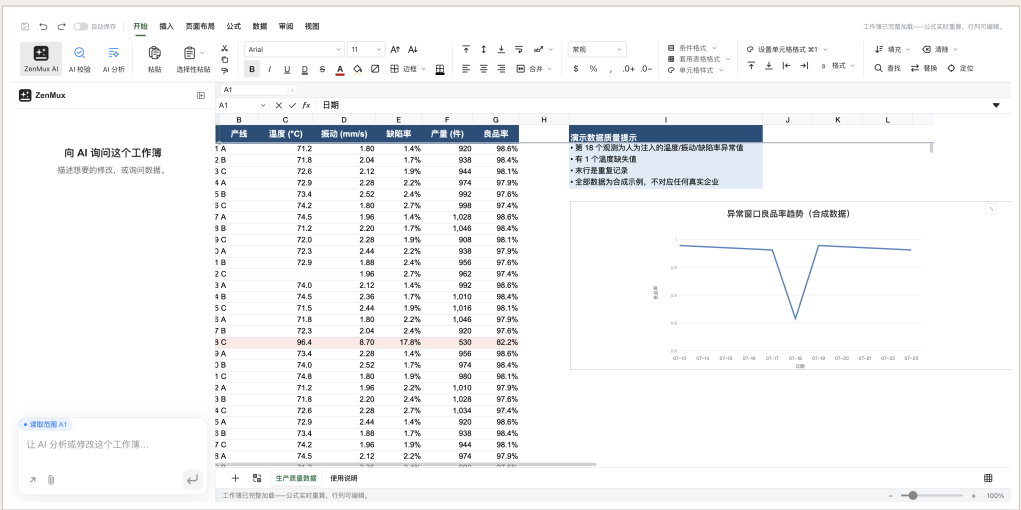
- 1. 0:00-0:30** In Word, open Insert > Equation > Insert new equation, then paste a capture or choose an image.
- 2. 0:30-1:30** Wait for ZenMux and verify sums, indices, inequality signs, matrix rows, and line breaks.
- 3. 1:30-2:30** Confirm both lines in preview, insert, select the new equation, and edit one symbol.
- 4. 2:30-3:00** Save, close, and reopen the DOCX to confirm that the equation is not a bitmap.

提示 • Tip If recognition is weak, crop away prose and keep only the equation. Never place unchecked OCR output in a final paper.

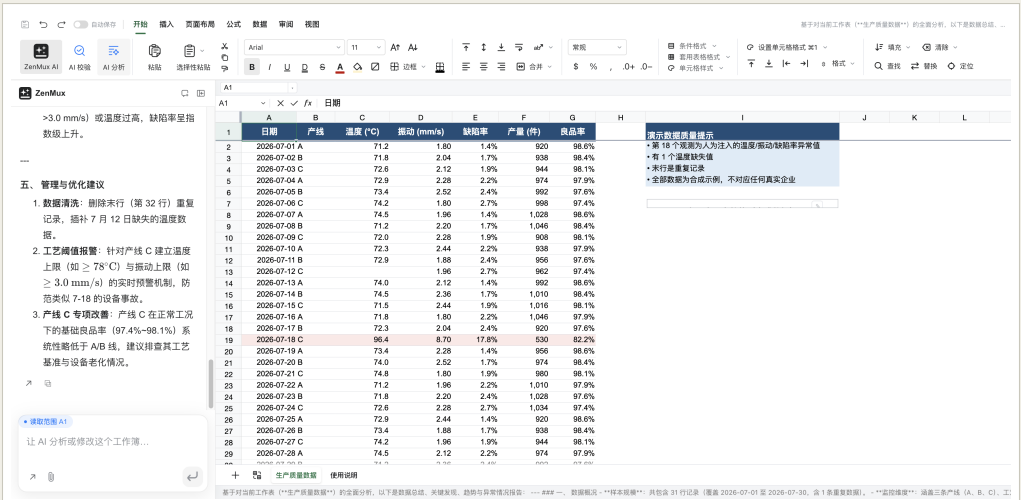
Recipe B: create a chart and EDA from an Excel table

场景 B：Excel 一键完成图表与 EDA

Open a real XLSX containing synthetic production-quality data, reveal the yield drop in a native chart, and ask ZenMux to find missing, duplicate, and jointly anomalous records.



Real runtime: editable data, a Chinese native line chart, and the injected anomaly in one view.



Real ZenMux analysis identifies missing data, the duplicate, anomaly thresholds, and improvement actions.

- 0:00-0:40** Open the XLSX and verify fields, units, and the synthetic-data notice.
- 0:40-1:30** Select date and yield, insert a line chart, and give it clear title and axis labels.
- 1:30-2:30** Choose AI Analysis and request missing values, duplicates, anomalies, trends, and evidence rows.

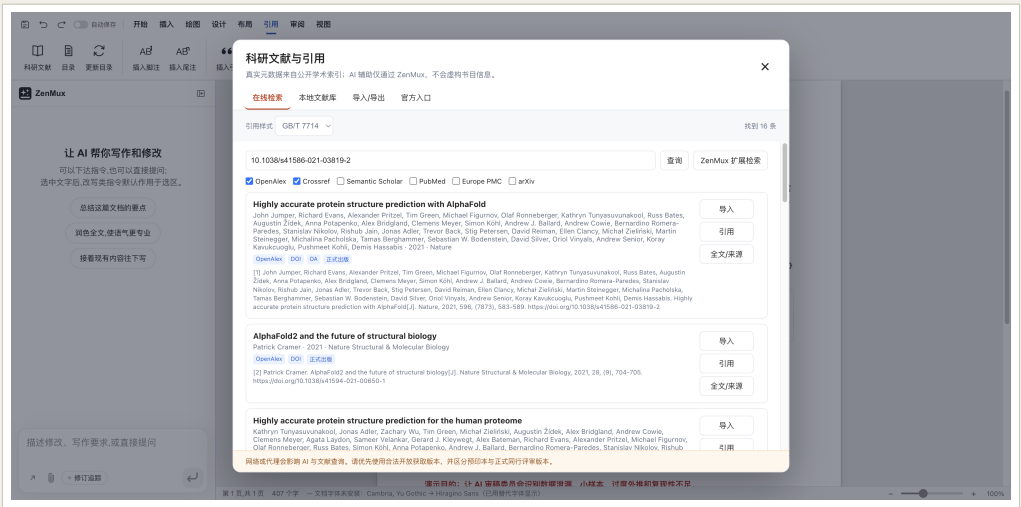
4. **2:30-3:00** Check observation 18: 96.4°C, 8.70 mm/s, 17.8% defects, and 82.2% yield. Reject any unsupported claim.

注意 • Notice This is deterministic synthetic data, not a real company dataset. Combine AI suggestions with business rules, statistical diagnostics, and expert review.

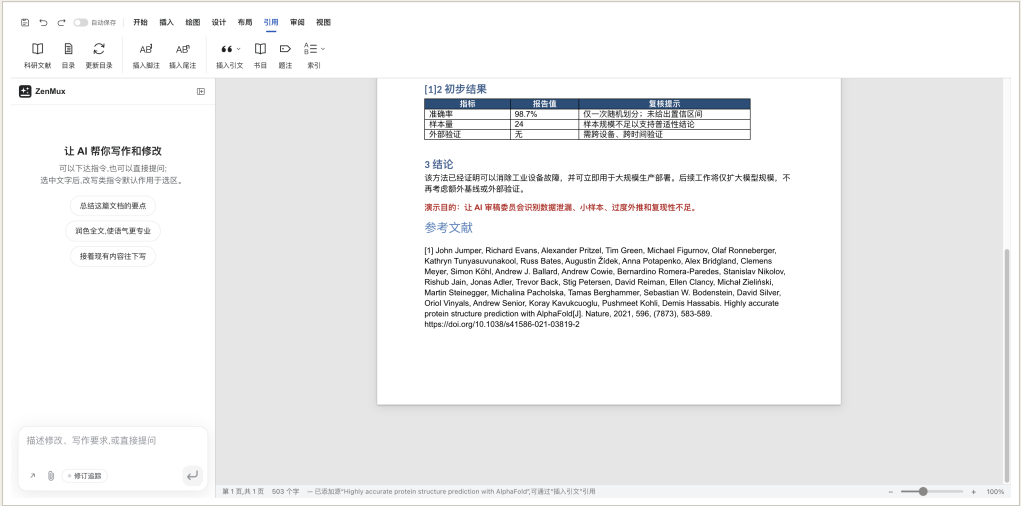
Recipe C: turn a DOI into a GB/T 7714 citation

场景 C：真实 DOI 变成 GB/T 7714 引用

Search public indexes for a real DOI, add the AlphaFold Nature article to the local library, insert an in-text citation, and generate the bibliography automatically.



Real query: OpenAlex/Crossref metadata, DOI/OA/version badges, and a GB/T 7714 preview.



Real write-back: References and the complete AlphaFold entry at the end of the document.

- 0:00-0:45** Open References > Research literature, choose GB/T 7714, and query 10.1038/s41586-021-03819-2.
- 0:45-1:30** Verify title, authors, Nature, 2021, volume/pages, and DOI, then import the published version.
- 1:30-2:20** Type /cite and choose the local entry, or use Cite beside the search result.

- 4. 2:20-3:00** Inspect the bibliography at the end, export .bib, and save DOCX and BIB together.

提示 • Tip Public metadata can still be incomplete. Verify against the DOI landing page or journal and distinguish preprints from peer-reviewed versions.

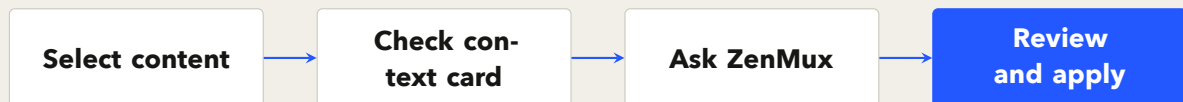
Select, ask, and route: the AI workflow

Five editors share context, attachments, clipboard input, equation rendering, message copy, and @Connect.

Use selected content as AI context

选择内容作为 AI 上下文

Select first, then ask. Word selects paragraphs, Excel selects cells, PowerPoint selects objects, Markdown selects text, and PDF captures a rectangular page region.



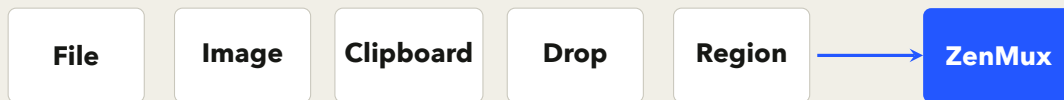
1. Make the selection and confirm that the UI shows a range, object, text card, or PDF region thumbnail.
2. State both action and boundary, for example: “Polish only the selected paragraph; preserve equations and citation keys.”
3. Read the response or diff before replacing, inserting, or applying. Undo and narrow the task if the result is not acceptable.
4. Markdown freezes the anchor at send time. If the source is edited later, a stale anchor is rejected to prevent misplaced write-back.

提示 • Tip With no explicit selection, AI may use the full document or current page. Explicit context reduces ambiguity, latency, and unnecessary input on long files.

Attachments, images, and clipboard

附件、图片与剪贴板

Each AI request accepts up to five attachments. Add files with the picker or drag-and-drop; pasted images appear as removable thumbnails and are sent as visual input.



- Ordinary files are parsed locally before relevant content is supplied; images are visual input. Large, encrypted, or unsupported files may not parse.
- Paste a clipboard image with Cmd/Ctrl+V. Check thumbnails before sending and remove any item that is not needed.
- If the total exceeds five, remove items or split the task. Never attach API Keys, passwords, or unredacted personal information.
- The app attempts to retain attachments and PDF regions after network failure. Confirm the thumbnail list again before retrying.

注意 • Notice Relevant attachment content is processed by the selected ZenMux model. Follow institutional rules before sending confidential, ethics-controlled, or otherwise restricted data.

Copy messages and restore file history

复制消息并恢复文档 AI 历史

Word, Excel, PowerPoint, Markdown, and PDF can copy both prompts and replies. Conversations persist locally by file path, and reopening the same file restores up to the latest 200 messages.

- Use the copy control beside a message. A plain-text destination receives source; a rich destination can preserve suitable structure.
- Save and close the document, then reopen the exact same absolute path. The AI panel shows “Earlier messages” ; restored items remain copyable and are returned to model context.
- History stays in local project storage. It is not written into document content, the GitHub repository, or an installer. Save As to a new path creates an independent history while the original path retains its conversation.
- Use $\$...\$$ for inline math and $\$ \$...\$ \$$ for display math. Multiline equations can use LaTeX environments such as `aligned` or `gathered`.
- GFM tables, lists, emphasis, and equations share the same reply parser. An unclosed delimiter remains readable during streaming and renders when complete.
- If a formula does not render, copy the source and check backslashes, paired delimiters, and accidentally escaped underscores.

Inline: $E = mc^2$ Multiline:
$$F = ma$$

$$E_k = \frac{1}{2}mv^2$$

Source remains copyable, rendering remains readable, and inserted equations remain editable.

提示 • Tip A wide table is a poor container for long reasoning. Ask for a concise table followed by prose to improve readability in Word and Markdown.

Cross-editor routing: send AI results with @Connect

跨编辑器联动：用 @Connect 一键流转 AI 结果

Type @Connect in an AI input to send the latest reply to another open Word, Excel, PowerPoint, or Markdown tab. The source can be any of the five editors, including PDF.



1. Open the destination first and confirm that its tab is visible in ZenOffice.
2. Type @Connect, then choose the correct item from the list of open targets. Do not rely only on similar filenames.
3. Verify the destination: Word/Markdown receives editable text, Excel receives cell-suitable content, and PowerPoint receives an editable text box rather than a flattened screenshot.
4. Transfer is local to the app, but the destination file still needs to be saved. If a target is closed or renamed, reopen it before selecting again.

提示 • Tip Useful flows: PDF region question → @Connect to Word for synthesis; Excel analysis → @Connect to PowerPoint for a briefing outline.

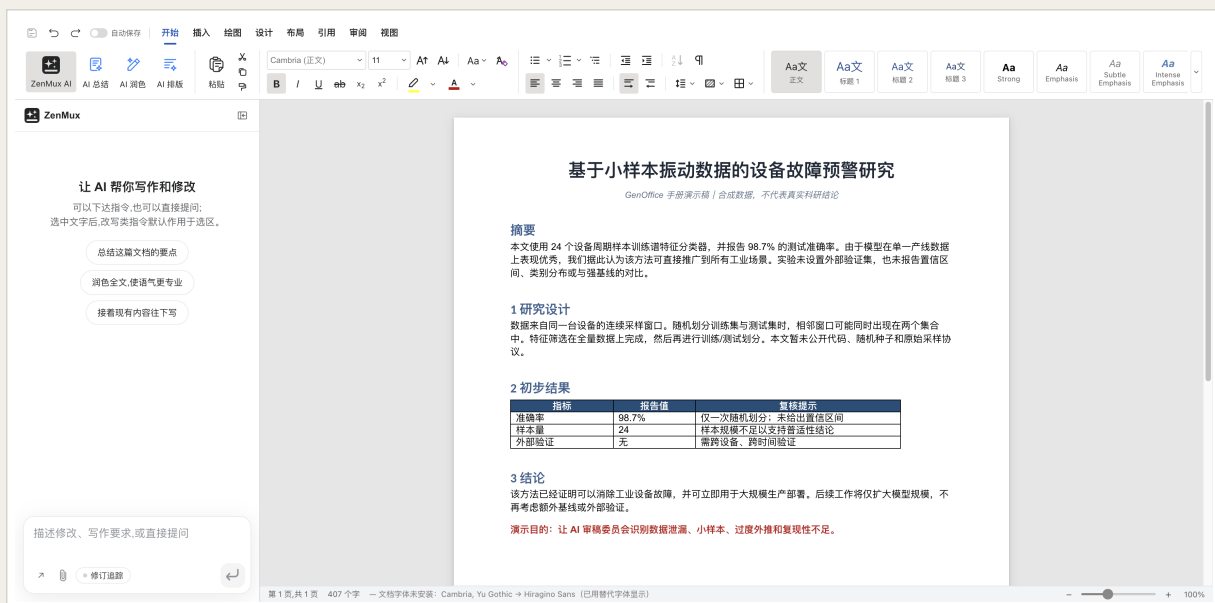
Word: from draft to pre-submission review

Edit real DOCX files with pagination, styles, image wrapping, equations, citations, review, and AI imagery.

Create, open, and save DOCX

创建、打开与保存 DOCX

The Word workspace edits real DOCX files. Unchanged areas are preserved as narrowly as possible, but page layout, fonts, images, and complex objects still deserve a final check.



Real runtime UI: Word editor, paginated canvas, ribbon, and the ZenMux AI panel on the left.

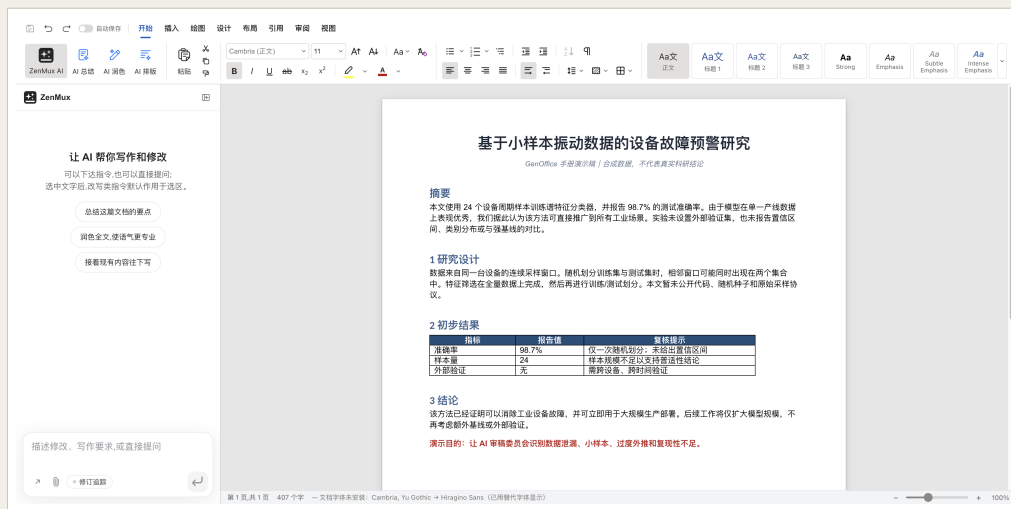
- Choose AI Docs to create a file or Open local file to select a .docx/.doc. The title area shows filename and save state.
- For legacy .doc, ZenOffice first asks local LibreOffice for a high-fidelity conversion to a separate DOCX, preserving fonts, sizes, colors, tables, and pagination. If LibreOffice is missing, the prompt can start the platform-appropriate installer: Homebrew on macOS, winget on Windows, and the native package manager with system authorization on Ubuntu/Debian, Fedora/RHEL, or Arch Linux. Unsupported environments fall back to the official download page. ZenOffice never degrades silently: text-only recovery runs only after explicit user confirmation. The original .doc is never overwritten.
- Use Cmd/Ctrl+S to save and Cmd/Ctrl+Shift+S for Save As. A new document asks for location and name on first save.
- Auto-save reduces accidental loss, but create a new version before major edits and respond to unsaved-close prompts.
- Review deliverables in Microsoft Word or another compatible app, especially custom fonts, fields, contents, floating objects, and multipage tables.

提示 • Tip Complex legacy layout, embedded objects, or macros may not convert completely. Keep the untouched DOC, converted DOCX, and final delivery as separate files.

Text, paragraphs, styles, and AI edits

文字、段落、样式与 AI 修改

The ribbon controls font, size, emphasis, color, alignment, lists, and styles. After selecting text, AI can polish, translate, summarize, or rewrite only that range.



Real Word runtime: editing surface beside the ZenMux panel, designed for select-then-ask work.

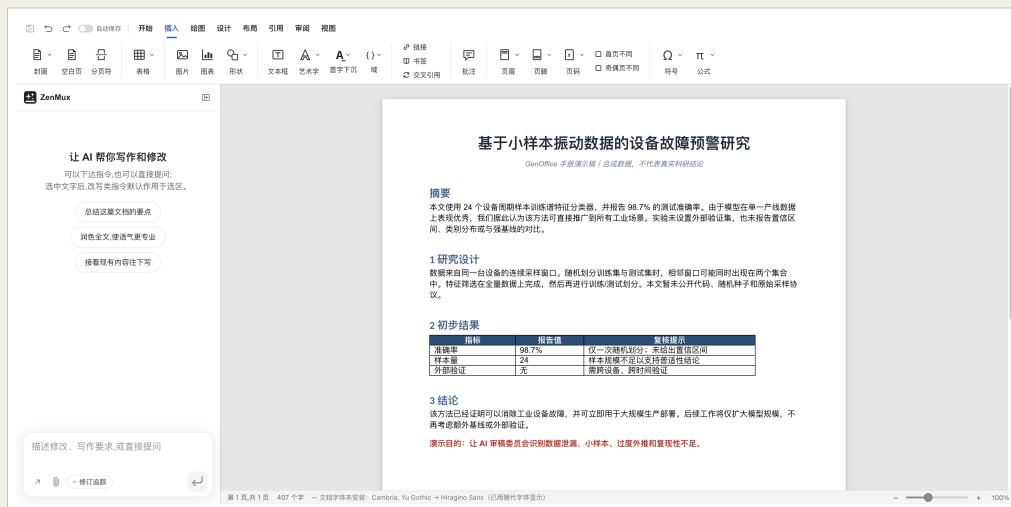
1. Select text before changing font or size. If the size control feels awkward, focus it, select the current value, type the target, and press Enter.
2. Build hierarchy with styles rather than manually enlarging every heading. Use paragraph controls for lists, indents, and spacing.
3. State language, tone, length, and protected items: “Preserve numbers, LaTeX equations, citation keys, and proper nouns.”
4. Review the diff before applying and test Undo afterward. Split a large multi-paragraph change into smaller batches to limit format side effects.

提示 • Tip Normalize styles before polishing a long manuscript. Content and layout changes then remain easier to review separately.

Insert images and wrap text

插入图片与文字环绕

Images can enter Word from a file, clipboard, or drag-and-drop. Layout modes include inline, square, tight, through, top and bottom, behind text, and in front of text.



Real runtime UI: Word Insert ribbon with image, table, equation, and drawing entry points.

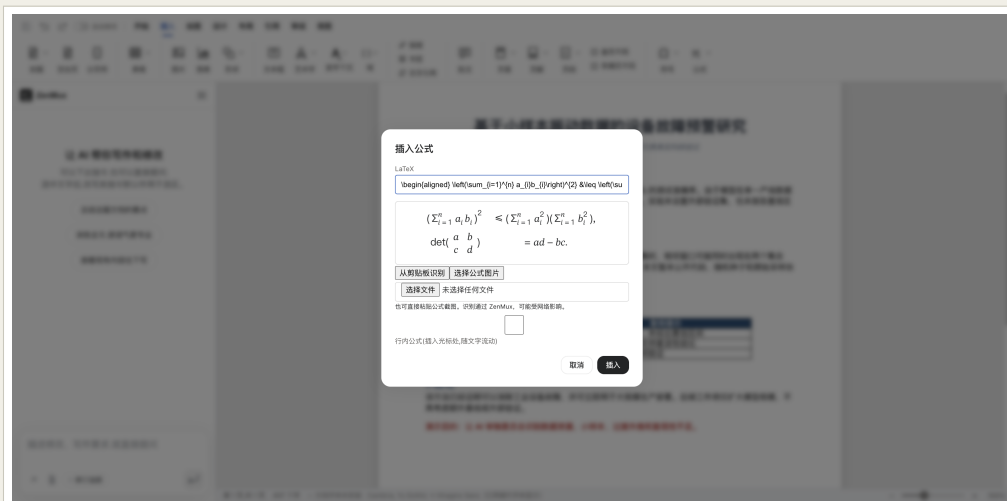
1. Choose Picture under Insert or drop a local image at the intended location. Confirm that the object remains inside the page boundary.
2. Prefer Inline for body content. Use square/tight for controlled floating layout; reserve Behind text for background-like material.
3. Resize proportionally from a corner. Width and height fields update the object; for floating images, verify anchor, z-order, and page position.
4. Before PDF export, compare paginated preview, paper size, margins, image dimensions, and borders. The exported page should match the canvas one-to-one.

注意 · Notice If an image appears to vanish in front-of-text or behind-text mode, check z-order, anchor page, page bounds, and zoom, then undo the most recent layout change.

Turn a screenshot into editable LaTeX

一张截图变成可编辑 LaTeX 公式

Word supports inline and display equations, multiline environments, and recognition from clipboard captures or image files. Confirm the recognized source before insertion.



Real ZenMux recognition: LaTeX source and rendered preview for the Cauchy inequality and a determinant.

- Inline math suits variables in a sentence, such as $F = ma$; display math suits derivations and numbered results. Use `aligned` for multiple lines:

$$F = ma,$$

$$E_k = \frac{1}{2}mv^2.$$

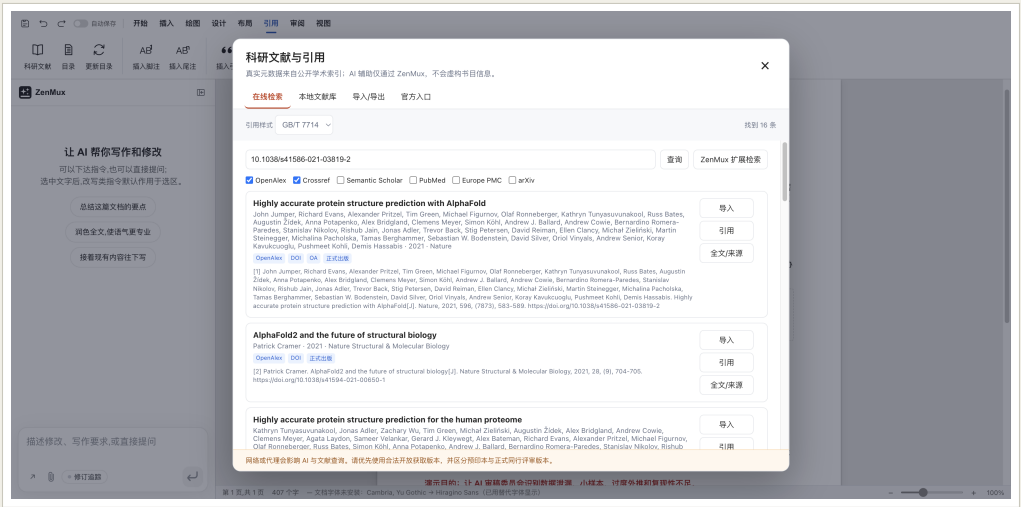
1. Enter LaTeX under Insert > Equation, preview, and insert. To revise, select the equation object before opening the editor.
2. For image recognition, paste a screenshot or choose a file. Crop to the formula and necessary number so background prose does not interfere.
3. Verify subscripts, superscripts, Greek letters, fractions, matrices, alignment points, and line breaks. Low-resolution or handwritten math may be misread.
4. Save and verify in Word that the result remains an editable equation rather than a screenshot.

提示 • Tip Keep a LaTeX source copy for complex formulas so the same math can be reused in Word, PowerPoint, Excel, and Markdown.

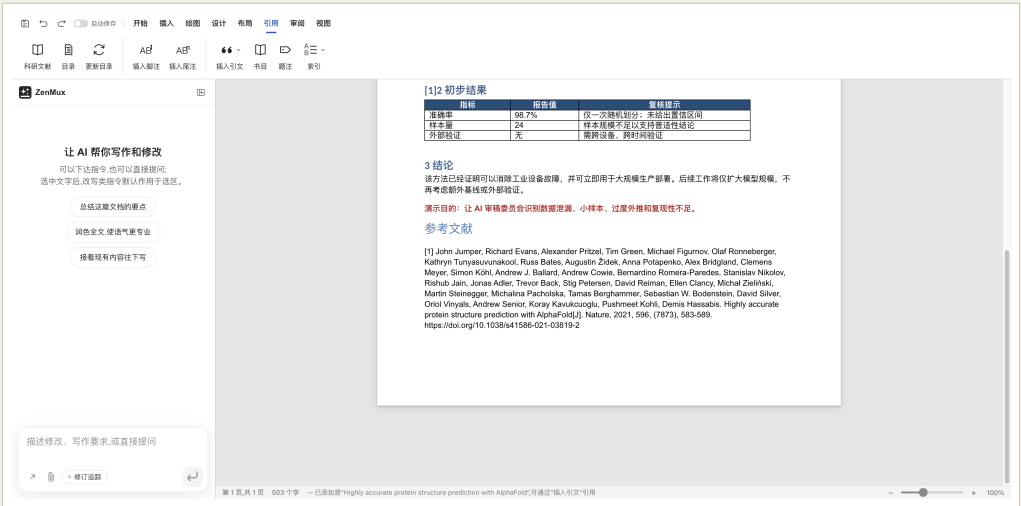
Find real literature and generate a complete citation

一键找到真文献并生成规范引用

Word References searches public scholarly metadata, imports a local library, inserts in-text citations, and generates both a bibliography and BibTeX.



Real DOI query: the AlphaFold Nature article, provenance badges, and GB/T 7714 preview.



Real write-back: a bibliography entry appears at the end and remains linked to the local library.

1. Search OpenAlex, Crossref, Semantic Scholar, Europe PMC/PubMed, and arXiv. Restricted services open official browser searches instead of being scraped.
2. Add results to the local library or import BibTeX, RIS, or CSL-JSON. Deduplicate by DOI, PMID, or arXiv ID and distinguish preprints from published versions.
3. Type `/cite` and choose from the local entry list, or insert through the manager. Select GB/T 7714, APA 7, IEEE, Nature, or Vancouver style.

4. Insert Bibliography when drafting is complete so entries appear at the end, then export .bib and keep even a temporary exchange file until delivery is verified.

注意 • Notice Public metadata does not imply that every full text can be downloaded lawfully. Prefer open-access copies and verify authors, title, year, journal, volume, pages, and DOI.

Reduce avoidable rejection: a three-reviewer committee plus chair

论文防拒稿利器：3 委员 + 1 主席严苛预审

Review creates three independent members plus a chair. Models are assigned from the available text-model list, and the report can be Chinese or English.



Real Science-level demonstration: the chair synthesizes three reports into a Reject decision and 5/5 confidence.



Real member output: independent concerns, questions for authors, and evidence for the verdict.

- Standards include Science, Nature, Cell, Elsevier, top IEEE journals, top IEEE conferences, general IEEE conferences, and several Chinese grant, 863, technology, and commercial proposal types.
- Members assess novelty, methods, evidence, statistics, reproducibility, writing, and compliance. The chair reconciles conflicts and separates major issues, minor issues, and recommendation.
- Scope includes prose, equations, tables, images, charts, and shapes. Visuals need adequate resolution plus complete captions, numbering, and textual references.

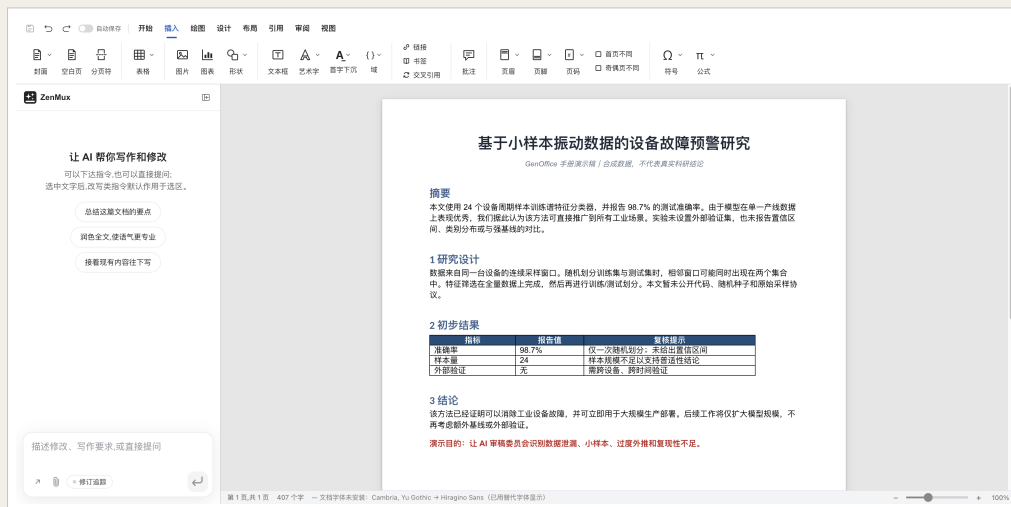
- Choose report language before starting. Long documents take time and depend on network conditions. Save the report and map every comment to a page, paragraph, equation, or figure.

注意 • Notice AI review is not formal peer review, grant adjudication, or legal advice. Random model assignment does not replace domain experts; verify every consequential conclusion.

AI imagery and scanned-page processing

AI 配图与扫描件处理

Word illustrations are generated through ZenMux and use a document-friendly aspect ratio. Scan tools can conservatively enhance monochrome material or remove handwriting over printed content.



Real Word Insert runtime: local images and AI illustrations enter the same document workflow.

1. Describe subject, composition, audience, aspect ratio, desired text, and prohibited elements. Word defaults to an illustration near a 3:2 document ratio.
2. For handwriting removal, select the scanned image and use `openai/gpt-image-2`. Instruct it to preserve print, equations, tables, lines, and paper structure.
3. Where handwriting overlaps print, reconstruction should be conservative and based only on nearby strokes. Enhancement mode is distinct from handwriting-removal mode.
4. Compare before and after at high zoom, checking digits, symbols, stamps, and edges. Replace only after confirmation and retain the untouched original.

注意 · Notice Image generation and restoration are uncertain and network-dependent. Never replace evidentiary, archival, signed, or original research records with an AI-processed image.

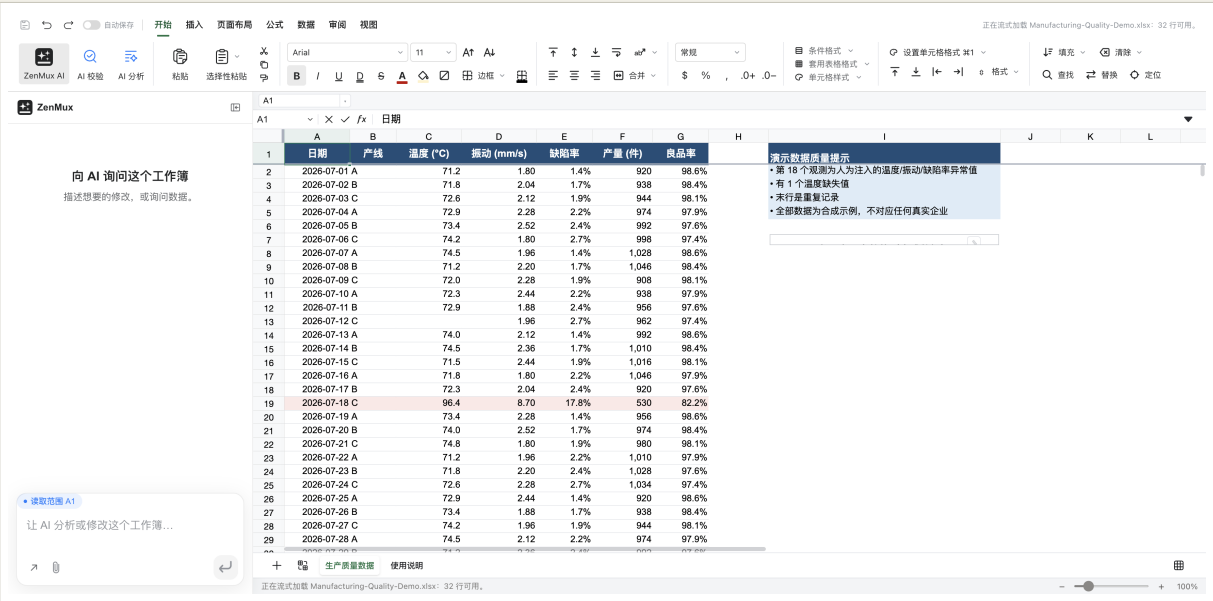
Excel: from raw data to defensible findings

Work in real XLSX files with formulas, cleaning, statistics, field-driven visualization, and selection-aware AI.

Workbooks, sheets, and basic editing

工作簿、工作表与基础编辑

The Excel workspace opens, edits, and saves real XLSX files with formula recalculation, sorting, filtering, validation, conditional formatting, pivots, and common workbook structure.



Real runtime UI: Excel grid, formula bar, ribbon, and ZenMux AI panel.

- Choose AI Sheets or open .xlsx/.xls/.csv. For CSV, verify encoding, delimiter, date conventions, and decimal separators.
- Select a range by mouse or name box; formulas begin with an equals sign. Confirm destination dimensions before pasting so adjacent data is not overwritten.
- Before sort/filter, use unique headers and avoid unnecessary merged cells. Validation is useful for categories, dates, and numeric limits.
- After saving, verify results in Excel or a compatible app, including formulas, charts, pivots, print area, and frozen panes.

提示 • Tip Duplicate the raw-data sheet and clean only the copy. Keep a data-dictionary sheet describing fields, units, missing values, and sources.

From raw data to defensible findings: complete EDA

从原始数据到可信结论：完整 EDA

After selecting existing fields, start with quality and descriptive statistics before correlation, regression, time series, grouped summaries, outliers, or scenarios.

The screenshot displays the ZenMux EDA interface. On the left, a sidebar lists analysis steps: 1. Data Cleaning (removing duplicates, filling missing values), 2. Process Monitoring (setting temperature and vibration limits), and 3. Line C Quality Improvement (analyzing defect rates). The main area shows a data table with columns: A1, A (Date), B (Line), C (Temperature), D (Vibration), E (Defect Rate), F (Production Volume), G (Yield Rate), and H (Data Governance提示). The table contains 29 rows of data. A tooltip for cell H1 provides details about data governance, including a warning about the 18th row being a manually entered temperature/vibration defect rate anomaly, a note about the 1st row being a temperature defect record, and a reminder to perform data governance checks.

A1	A	B	C	D	E	F	G	H
	日期	产线	温度 (°C)	振动 (mm/s)	缺陷率	产量 (件)	良品率	
1	2026-07-01	A	71.2	1.80	1.4%	920	98.6%	显示数据治理提示 • 第 18 个数据为人为注入的温度/振动缺陷率异常值 • 第 1 个温度缺陷记录 • 未行重复记录 • 全部数据为合成示例，不对任何真实企业
2	2026-07-02	B	71.8	2.04	1.7%	938	98.4%	
3	2026-07-03	C	72.6	2.12	1.9%	944	98.1%	
4	2026-07-04	A	72.9	2.28	2.2%	974	97.9%	
5	2026-07-05	B	73.4	2.52	2.4%	992	97.6%	
6	2026-07-06	C	74.2	1.80	2.7%	998	97.4%	
7	2026-07-07	A	74.5	1.96	1.4%	1,028	98.6%	
8	2026-07-08	B	71.2	2.20	1.7%	1,046	98.4%	
9	2026-07-09	C	72.9	2.28	1.9%	908	98.1%	
10	2026-07-10	A	72.3	2.44	2.2%	938	97.9%	
11	2026-07-11	B	72.9	1.88	2.4%	956	97.6%	
12	2026-07-12	C	73.4	1.96	2.7%	962	97.4%	
13	2026-07-13	A	74.0	2.12	1.4%	992	98.6%	
14	2026-07-14	B	74.5	2.36	1.7%	1,010	98.4%	
15	2026-07-15	C	71.5	2.44	1.9%	1,016	98.1%	
16	2026-07-16	A	71.8	1.80	2.2%	1,046	97.9%	
17	2026-07-17	B	72.3	2.04	2.4%	920	97.6%	
18	2026-07-18	C	96.4	8.70	17.6%	530	82.2%	
19	2026-07-19	A	73.4	2.28	1.4%	956	98.6%	
20	2026-07-20	B	74.0	2.52	1.7%	974	98.4%	
21	2026-07-21	C	74.8	1.80	1.9%	980	98.1%	
22	2026-07-22	A	71.2	1.96	2.2%	1,010	97.9%	
23	2026-07-23	B	71.8	2.20	2.4%	1,028	97.6%	
24	2026-07-24	C	72.6	2.28	2.7%	1,034	97.4%	
25	2026-07-25	A	72.9	2.44	1.4%	920	98.6%	
26	2026-07-26	B	73.4	1.88	1.7%	938	98.4%	
27	2026-07-27	C	74.2	1.96	1.9%	944	98.1%	
28	2026-07-28	A	74.5	2.12	2.2%	974	97.9%	
29	2026-07-29	B	74.5	2.12	2.2%	974	97.9%	

Real ZenMux EDA identifies the joint anomaly, missing value, duplicate row, and actionable data-governance steps.

Define fields/units

Check missing/outliers

Choose method

Validate/interpret

1. Define the question and unit of analysis. Confirm what each row represents, column types, sample scope, time frequency, and missing-value codes.
2. Run quality checks and descriptive statistics before choosing correlation, covariance, regression, group comparison, trends, forecasting, or anomaly detection.
3. Place output in a new range or sheet. Preserve raw data and record filters, parameters, and any randomness.
4. Test assumptions, sample size, outliers, and leakage. Correlation is not causation; intervals are more informative than a single forecast point.

注意 • Notice AI can suggest methods and generate steps, but it does not replace research design, statistical diagnostics, or domain judgment. Obtain expert review for consequential conclusions.

Make the data speak with an editable chart

让数据说话：一键生成可编辑中文图表

Excel can recommend and create editable native charts from selected columns, with Chinese-friendly titles, axes, legends, labels, and hover values.



Real runtime close-up: the Chinese title and axes remain legible, observation 18’s anomalous trough is clear, and the chart stays editable.

Question	Suitable chart
Category comparison	Column, bar, or combo
Trend over time	Line, area, or combo
Part-to-whole	Pie or doughnut, with few categories
Relationship	Scatter, optionally with a trend line
Multivariate profile	Radar, with careful scaling

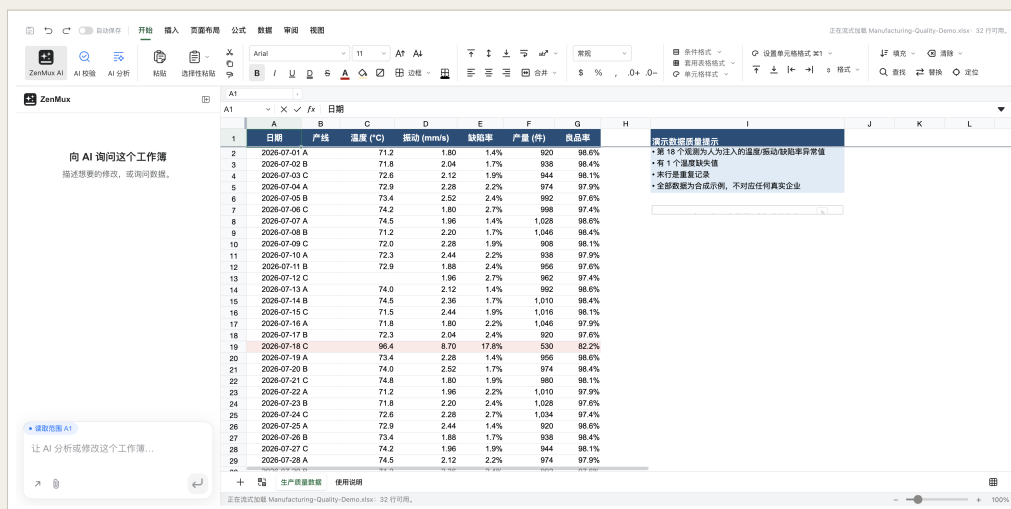
- Select a range with headers and verify that category and numeric columns are aligned.
- Edit title, axes, legend, color, labels, and data source. Ensure chosen Chinese fonts exist on the delivery system.
- Hover provides lightweight interaction, but static exports still need visible titles, units, annotations, and source notes.

提示 • Tip Let one chart answer one primary question. Several focused charts are usually clearer than one overloaded figure.

Analyze a selected range and write back safely

选中一块数据，让 AI 分析并安全回填

After selecting a cell range, AI can explain data, generate formulas, clean or format, and write back. Validate every batch operation on a small range first.



Real Excel runtime: a grid selection can become precise ZenMux AI context.

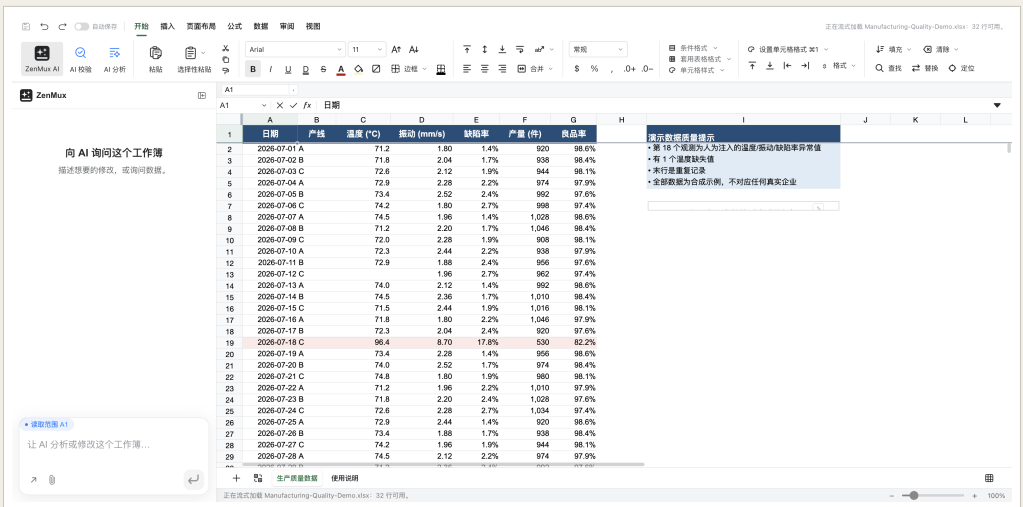
1. Select the target and state whether headers are included, which columns may be added, the first formula row, and how blanks should be treated.
2. Ask for a plan and preview before applying. Test complex formulas on a few rows and verify relative versus absolute references.
3. Recognize an image formula from clipboard or file as LaTeX, then convert it to a spreadsheet formula where appropriate or insert it as explanatory math.
4. Check number formats, percentages, dates, alignment, conditional formatting, and fill range. Neutral alignment such as right-align should map safely without aborting the batch.

提示 • Tip Creating a new result column beside the source is safer than overwriting it. Replace only after verification.

Model a workbook as a relational database

把一个工作簿定义成关系数据库

Under Data > SQL Database, one workbook becomes a local database and every visible worksheet becomes a table. The schema panel turns ordinary headers into validated field definitions.



Open Data > SQL Database from the Excel workspace; the workbook remains an editable, saveable XLSX file.

Schema item	Usage
Table and field names	Sheet names become tables and the first row becomes fields. Wrap Chinese or spaced identifiers in square brackets.
Field types	Choose TEXT, INTEGER, NUMBER, BOOLEAN, DATE, or DATETIME and set nullability.
Primary key	Select one field for a simple key or several for a composite key. Primary-key fields cannot be NULL.
Secondary index	Build ordinary or unique indexes over one or several fields; multiple fields form a composite index.
Local persistence	Choose Save schema & rebuild. Schema and keys persist locally; work-sheet data is not uploaded.

注意 • Notice This database is a local query view of the current workbook, not a remote server. After structural row/column changes, save the workbook and reopen the SQL workspace.

SQL workbench: script, diagnose, and write results

SQL 工作台：写脚本、定位错误并回填结果

The workbench combines a table catalog, SQL editor, results, and schema inspector for reproducible joins, aggregation, and filtering across worksheets.



- From v0.6.54, the workbench uses a light interface. Its ZenMux AI area provides a resizable multiline prompt, reads the real catalog, generates and verifies read-only SQL, and can insert the returned SQL into the editor.
- SQL AI follows the same attachment rules as the Word, Excel, PowerPoint, Markdown, and PDF assistants: up to five files per request, added through the picker or drag-and-drop, plus clipboard images. Images appear as removable thumbnails; other files show their names and sizes.
- The editor provides syntax highlighting, line numbers, multi-statement scripts, folding, and in-line diagnostics. **Cmd/Ctrl+Enter** runs the selection or current statement; **Shift+Cmd/Ctrl+Enter** runs the full script.
- Errors point to the relevant line and statement; numbered tabs switch among multiple results. Preview a SELECT result, then Write to Excel to create a **new worksheet** without overwriting sources.
- Safe read-only mode rejects mutations by default. Enabling session writes changes only the in-memory database and never writes them automatically to XLSX.
- AI must load the real catalog before generating, explaining, or repairing SQL. Automatic execution is restricted to SELECT, WITH, or EXPLAIN; a result is inserted only after an explicit user request. A failed network run retains its prompt and attachments for retry; verify the attachment strip before resending.

提示 • Tip Verify join conditions and row counts with SELECT before writing back. Use square brackets for Chinese identifiers, for example [订单明细].[商品 ID].

Download two sample databases and reproduce the tests

下载两个样例数据库并复现实测查询

The Landing Page hosts two ready-to-open XLSX samples. Hidden _GenOfficeSchema metadata restores types, keys, and indexes, and both files pass the real XLSX reader plus SQL test suite.

Sample		Model and validation focus
Retail database Research database	Orders	Customers, orders, lines, and products; validates 1:N relations, composite line key, discounts, status, and missing values.
	Projects	Projects, researchers, papers, and membership; validates M:N relations, composite membership key, budgets, dates, and missing ORCID values.

Retail sample: sales by city for non-cancelled orders

```
SELECT c.[城市], ROUND(SUM(d.[行金额]), 2) AS [销售额]
FROM [客户] c JOIN [订单] o ON o.[客户 ID] = c.[客户 ID]
JOIN [订单明细] d ON d.[订单 ID] = o.[订单 ID]
WHERE o.[订单状态] <> '已取消' GROUP BY c.[城市]
ORDER BY [销售额] DESC;
```

提示 · Tip Open the XLSX → Data → SQL Database → paste the query → Cmd/Ctrl+Enter → Write to Excel. Use the research sample for GROUP BY, HAVING, date aggregation, and many-to-many joins.

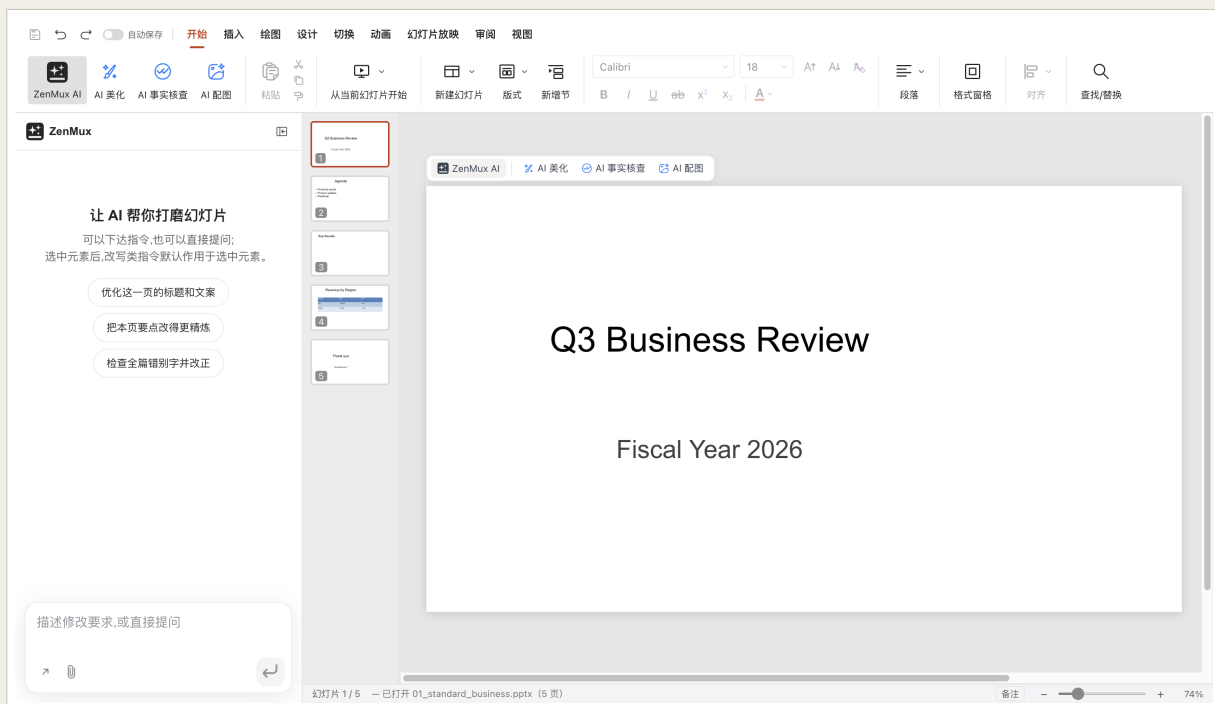
PowerPoint: build a genuinely editable deck

Generate and edit real PPTX objects with shapes, equations, imagery, media, recording, and editable output.

Open, edit, and generate editable slides

打开、编辑与生成可编辑幻灯片

The presentation workspace edits PPTX directly. AI output should become editable text boxes, shapes, tables, and charts rather than one flattened image per slide.



Real runtime UI: slide canvas, thumbnails, ribbon, and ZenMux AI panel.

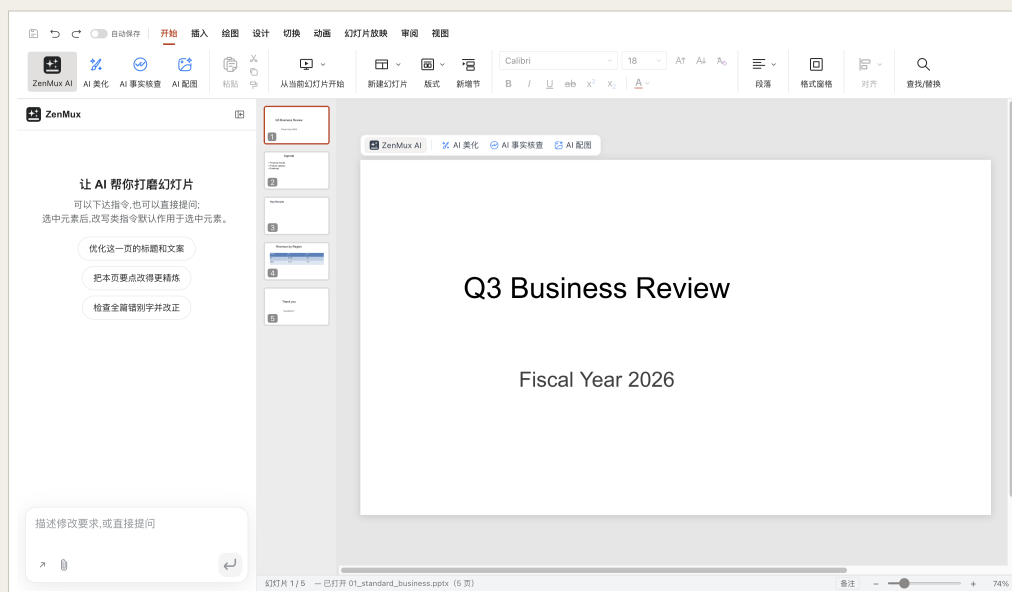
- Open a .pptx or create one with AI Slides. Define aspect ratio, theme, master, and audience before generating content.
- Newly generated slides use projection-readable type: 44–54pt cover titles, 34–42pt page titles, and 21–28pt body copy. Body slides normally keep three to five points; dense material is tightened or split rather than forced into tiny type.
- Double-click a text box to edit. Select an object to change position, size, fill, outline, font, and z-order; smart guides assist alignment.
- Explicitly request editable objects and prohibit full-slide flattening. Limit the number of ideas on each slide.
- After saving, review in PowerPoint or a compatible app, including masters, fonts, charts, animations, media, and object bounds.

提示 • Tip Generate a five-to-eight-slide outline first, then refine each page. This is easier to control than producing a long deck in one request.

Shapes, WordArt, tables, and charts

形状、艺术字、表格与图表

The Drawing tab includes 104 editable shapes covering common Visio-style basics. WordArt, tables, and charts should remain consistent across canvas, editing, and export.



Current presentation runtime: object canvas and ribbon for selection, alignment, grouping, and z-order.

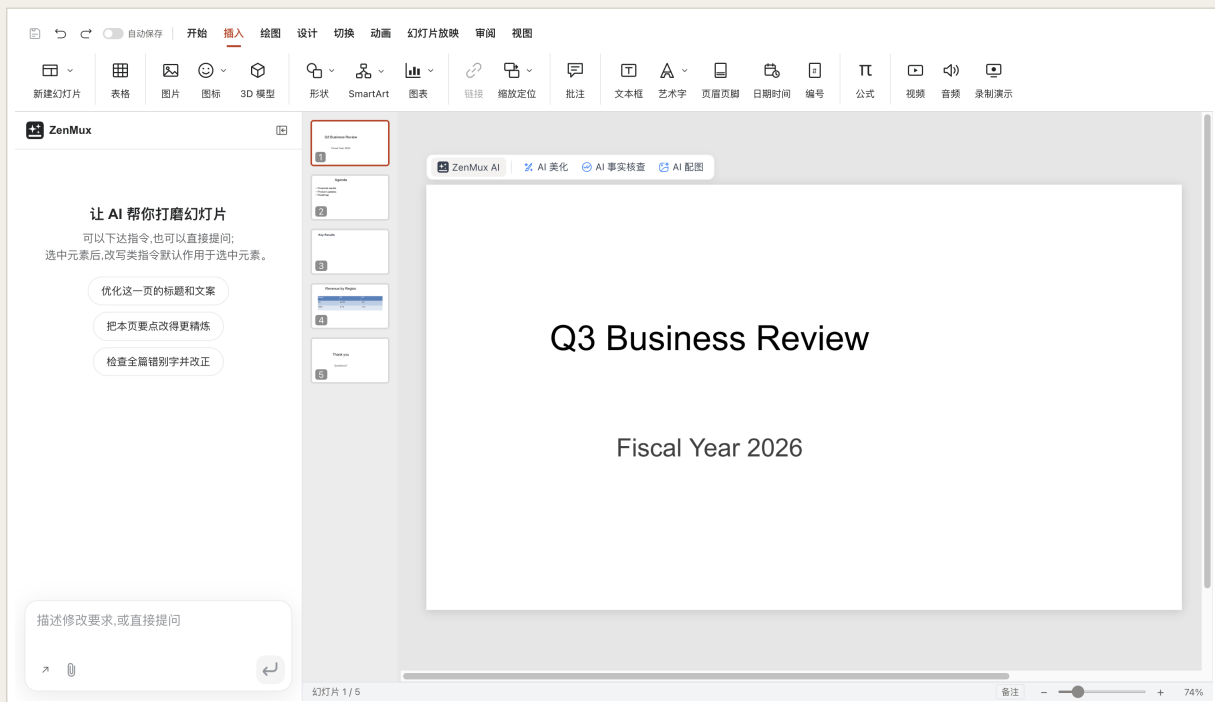
1. Insert rectangles, circles, arrows, flowchart symbols, callouts, stars, and connectors from the shape library; use modifier keys to constrain or duplicate.
2. Normalize fill, outline, corners, and typography. Use align, distribute, group, and z-order rather than estimating by eye.
3. For WordArt, verify text, font, size, fill, true outline, rotation, and scaling—not only the on-screen appearance.
4. Table cells remain editable; chart data, legends, axes, and labels should remain updateable. Run a slide show to find objects outside the page.

提示 • Tip Build a process diagram from shapes and connectors instead of a screenshot so users and AI can revise nodes and labels later.

Insert and play video, audio, and images

插入并播放视频、音频和图片

Drop video, audio, or images directly onto a target slide, or choose a file under Insert. Media plays inside its inserted frame in both edit preview and slide show.



Real runtime UI: PowerPoint Insert ribbon with image, media, equation, and object entry points.

- Common video containers include MP4, M4V, MOV, and WebM. Actual playback still depends on system codecs and the media encoding.
- Drop at the intended location and resize from a corner. Playback should fit the inserted frame while preserving aspect ratio, without horizontal or vertical distortion.
- Test start, pause, end, volume, and slide transitions in both edit preview and slide show; a static poster frame is not enough.
- Images can also be dropped and remain movable, croppable, and resizable. Large media increases both PPTX and exported-video size.

提示 • Tip Test a delivery copy on another target-platform machine to catch broken media references, missing audio, or aspect-ratio differences.

Record narration and export MP4

录制旁白并合成 MP4

Recording is available from Slide Show and Insert. ZenOffice captures microphone narration and composes the presentation into a local H.264/AAC MP4 with pause and resume.

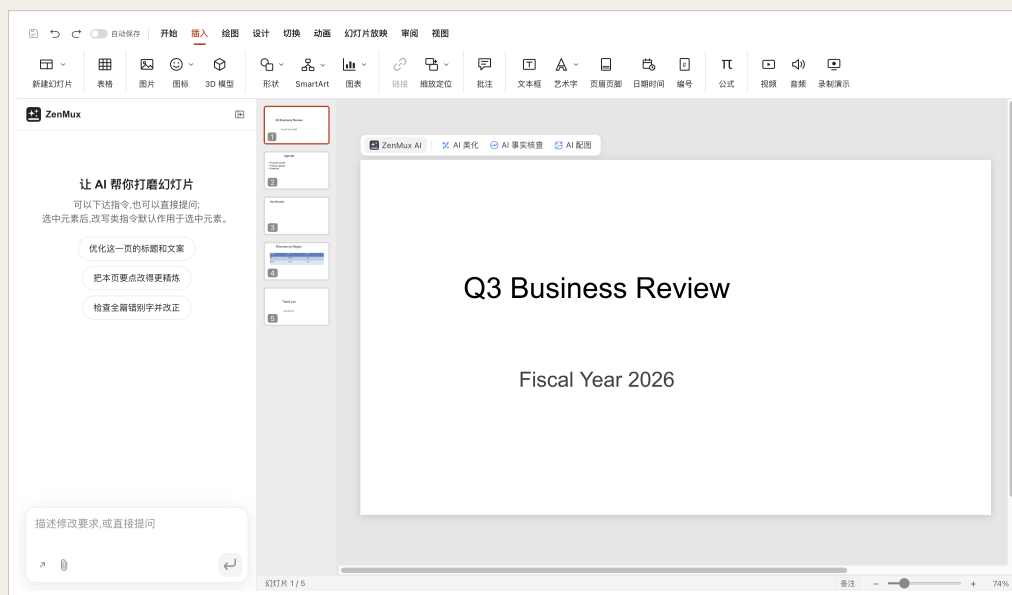
1. Before the first recording, grant microphone access. macOS may also require Screen Recording permission; quit and relaunch the app after changing permissions.
2. Choose the microphone and watch input activity. Record a ten-second test and check for silence, clipping, and intelligibility.
3. macOS mixes the microphone with media played inside the ZenOffice presentation. Windows can use system loopback. System-wide audio availability depends on platform and permission support.
4. Start the full-screen show, pause/resume as needed, wait for encoding, choose a local MP4 destination, then verify image, sync, aspect ratio, and duration.

注意 • Notice If narration works but system audio is missing, distinguish slide-embedded media from audio in other apps, then check platform capability, screen/audio permissions, and output device.

Equations, citations, and imagery

PPT 公式、文献与 ZenMux 配图

PowerPoint supports LaTeX equations, image-to-formula recognition, scholarly search and citations, plus ZenMux imagery composed for 16:9 slides.



Real Insert runtime: equations, images, and media enter the current slide as layout-aware objects.

- Enter or paste LaTeX under Insert > Equation. Preview multiline derivations before placing them in a display box; verify every recognized symbol.
- Literature search and the local library are shared with Word. Use editable in-slide citations, place references on final slides or in notes, and export .bib.
- An illustration prompt should state visual focus, desired empty space, brand colors, and 16:9 landscape composition. Crop without distorting aspect ratio.
- Credit external images, data, and papers in the footer. Verify dates, authors, and original URLs for web-derived claims.

提示 • Tip In scientific presentations, use editable charts and shapes for evidence. Generated imagery should support explanation or mood, not substitute for real data.

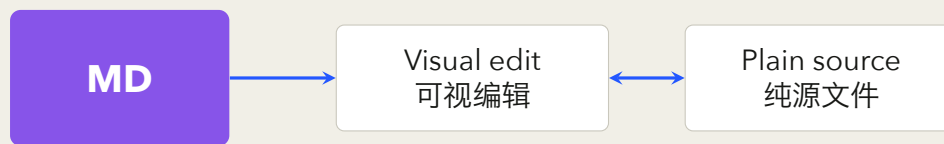
Markdown & PDF: structured writing and reading

Manage editable Markdown, equations, and diagrams; use PDF for reading, page operations, watermarks, security, and visual questions.

Markdown basics and images

Markdown 基础编辑与图片

The Markdown workspace round-trips plain .md/.markdown files with headings, lists, tasks, quotations, GFM tables, links, code blocks, and local images.



- Create AI Markdown or open .md. Saving writes plain Markdown rather than a proprietary binary format.
- Use the ribbon or familiar syntax for headings, lists, tasks, quotes, links, tables, and fenced code blocks.
- Insert images by picker, paste, or drop. Prefer paths relative to the .md file and move the asset folder with the document.
- Select text to ask, rewrite precisely, or insert AI output before/after the anchor. Confirm that the context card still matches before applying.

提示 • Tip For a portable Markdown project, keep the .md, image folder, and .bib under one parent directory and use relative paths.

Inline and multiline math in Markdown

Markdown 中的行内与多行 LaTeX

Markdown supports inline, display, and aligned multiline math plus recognition from clipboard or image files. Correct delimiters and backslashes are essential.

Inline source: `$F_1 = G \sin\theta$` Rendered: $F_1 = G \sin \theta$

Rendered multiline:
$$F = ma$$
$$E_k = \frac{1}{2}mv^2$$

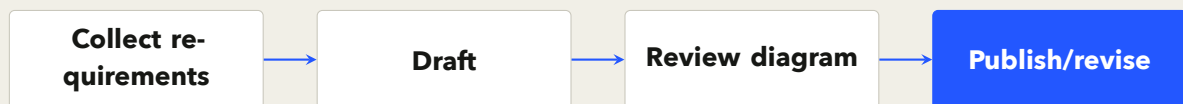
- Use ASCII dollar signs, not full-width variants. Opening and closing `$$` for display math must be paired.
- Preserve command backslashes. Write a subscript as `F1`; avoid turning commands into over-escaped plain text.
- Use `\\` for line breaks and close every matrix, cases, aligned, or other environment.
- After image recognition, inspect LaTeX source before insertion. Save and reopen to verify both source and rendering.

提示 • Tip If a large pasted passage fails to render, inspect plain source for unmatched dollar signs, missing backslashes, and incorrectly escaped underscores.

Insert, edit, and generate Mermaid

Mermaid 插入、编辑与 AI 生成

Markdown supports standard fenced Mermaid source, live rendering, source edits, and ZenMux generation or modification from natural-language instructions.



1. Insert a Mermaid fence and choose the right diagram family: flowchart, sequenceDiagram, stateDiagram, classDiagram, and so on.
2. Quote node labels that contain @, parentheses, colons, or other special characters, for example A["Label"].
3. Check syntax and then logic—direction, missing nodes, cycles, and sensitive information. Ask AI to return the complete fenced block when modifying.
4. Save and reopen the .md to verify rendering. During export or Word conversion, confirm that the diagram is preserved appropriately.

注意 • Notice A Mermaid diagram that parses can still be logically wrong. A domain owner must validate process, state, and sequence relationships.

Convert between Markdown and Word

Markdown 与 Word 互转

Markdown can export to Word, and Word content can be converted to Markdown. The goal is editable structure, not pixel-identical layout between two different models.



- Before Word export, use proper heading levels, real lists, GFM tables, relative image paths, and closed math delimiters.
- Verify multiline math after conversion to Word equations. Mermaid may become a graphic or alternate representation; check the exported result.
- Word-to-Markdown cannot map every pagination feature, floating image, WordArt object, header/footer, or style. Simplify manually where necessary.
- Keep both source and result. Choose one as the new master instead of alternating edits and creating divergent versions.

提示 • Tip Markdown suits version control and lightweight collaboration; Word suits complex pagination and formal delivery. Choose the master according to the final use.

Markdown review, translation, and literature

Markdown 审稿、翻译与文献

Markdown shares Word's strict review path, selection/full-document translation, and scholarly library, including in-text citations, end bibliography, and .bib export.

1. Choose a target standard and Chinese or English report. The committee reviews prose, equations, tables, images, and Mermaid diagrams.
2. During translation, protect code fences, links, citation keys, LaTeX, and Mermaid source. Process long files by section and maintain a glossary.
3. Search or import literature, then insert Pandoc-compatible keys from the local library. Type `/cite` to choose an entry.
4. Generate references at the end and export BibTeX. Inspect plain source to confirm that keys, bibliography block, and image paths were actually saved.

提示 • Tip Before translating, define terminology, abbreviations, and names that must remain unchanged. Compare against the source paragraph by paragraph afterward.

PDF viewing and page operations

PDF 预览与页面处理

The PDF workspace offers continuous, single-page, and two-page views; zoom, fit page/width, hand pan; and page extraction, insertion, rotation, reorder, and deletion.



- Open through AI PDF, Open local file, an operating-system association, or the dedicated PDF entry point. Encrypted files request an open password.
- Switch between continuous, single, and spread views. After zooming, use the hand tool to pan without selecting annotations accidentally.
- Save a copy before page operations. After extract, insert, rotate, reorder, or delete, check page numbers, bookmarks, forms, signatures, and total count.
- The current PDF ribbon does not expose text editing, image editing, or equation insertion. For content-level changes, edit the source document and export PDF again.

注意 · Notice Deleting or reordering pages changes cited page numbers and may invalidate signatures. Check compliance before modifying contracts, archives, or signed PDFs.

PDF watermarks, annotation, and security

PDF 水印、批注与安全

Watermark text accepts clipboard paste and configurable color, angle, opacity, and size. After confirmation and save, it is embedded on every PDF page rather than left as preview-only UI.

1. Open Watermark and paste text. Pasting over selected text replaces the selection; multiline input is normalized into a suitable single-line watermark.
2. Preview color, angle, opacity, and size, ensuring that key text, signatures, QR codes, and page numbers remain readable.
3. Confirm and save a copy, then close and reopen it in another PDF reader to verify every page contains the watermark.
4. Annotations, forms, signatures, stamps, headers/footers, and automatic page numbers are available. Save encryption as a new AES-128 protected copy rather than overwriting the source.

注意 · Notice The application does not persist PDF passwords. A lost password can make the protected copy inaccessible; store it separately with an approved password manager.

PDF region questions, attachments, and @Connect

PDF 框选提问、附件与 @Connect

Drag a rectangle over a page to make that capture the primary visual context. PDF AI also supports up to five attachments, pasted images, prompt/reply copy, history restored by PDF path, and @Connect.



1. Activate region capture and drag on the current page. Inspect the thumbnail at high zoom so text, table, figure, or equation is complete; press Escape to cancel.
2. Add files by picker/drop or paste images, up to five total. Ordinary files are parsed locally; images are visual input.
3. State whether to recognize, explain, verify, or transcribe. For tables and equations, require units, footnotes, symbols, superscripts, and subscripts to be preserved.
4. Copy either the prompt or reply, or use @Connect to send the reply to an open Word, Excel, PowerPoint, or Markdown tab. Reopening the same PDF path restores up to the latest 200 messages.

提示 • Tip Before retrying a failed network call, confirm that the region thumbnail and attachments remain. On a multicolumn page, capture one logical region at a time.

Web verification, reliability & delivery

Ground AI in the current date and credible sources while mastering retries, permissions, recovery, privacy, and delivery checks.

Make “latest” current: verify against today’s date

让“最新”真的最新：按当前日期联网查证

The system date when this manual was updated is 29 August 2026. Questions about “latest,” “today,” current policy, markets, or recent science should explicitly request current-date web verification.

1. Define a time boundary: “Use the current system date as the cutoff and rely only on verifiable material after 1 January 2025.”
2. Ask the response to distinguish model knowledge, retrieved facts, and inference/forecast, with source title, institution, date, and original URL.
3. Cross-check news with sources such as Reuters/AP, BBC, CNN, or Times. Use X to discover leads, not as the sole evidence.
4. For science, prefer original papers, Science, Nature, PubMed/PMC, societies, universities, and government databases. Label preprints as not formally peer reviewed.

注意 • Notice Web pages change and search snippets omit context. For high-stakes claims, open the original, record access date, and use at least two independent credible sources.

Scholarly sources and lawful full text

科研检索入口与合法全文

Scholarly search combines open APIs with a local library; other services open their official search pages. Prefer lawful open access and never bypass controls or scrape restricted services.

Need	Recommended entry points
General metadata	OpenAlex, Crossref, Semantic Scholar
Biomedicine	PubMed, PMC, Europe PMC
Preprints	arXiv, bioRxiv, medRxiv, SSRN, OSF
Open full text	CORE, DOAJ, BASE, repositories, Unpaywall
Chinese discovery	Baidu Scholar, PubScholar, NSSD, NSTL under official access rules

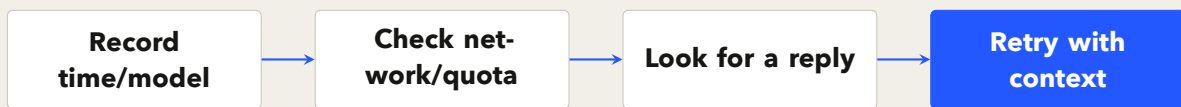
- Search with DOI or a combination of title, author, and year. Google Scholar is useful for discovery, but the app should open official search rather than scrape it.
- Check license, OA status, and version before download. Accepted manuscripts, preprints, and versions of record can differ in content and review status.
- Deduplicate the local library and complete type, pages, and DOI. Cite the formal version where possible and label a preprint when used.

提示 · Tip Unpaywall and Open Access Button help locate lawful versions. Institutional subscriptions and document delivery remain subject to their license terms.

Recover safely from timeout or interrupted streaming

AI 超时也不丢结果：判断状态并安全重试

A client timeout does not necessarily mean the provider produced no result. Do not immediately resubmit the same long task repeatedly; determine request state and preserve context first.



1. Record send time, model, prompt length, and attachment count. Wait briefly and see whether streaming resumes in the current conversation.
2. Check ZenMux status, quota, network, and proxy. Use a short prompt or built-in model as a health test.
3. If the provider console shows completion but the app did not write back, copy any available result or retry in the original conversation and retain the prompt for diagnosis.
4. Confirm that the selection, PDF region, and up to five attachments remain. Split long work into plan, sectional generation, and merge phases.

注意 • Notice Repeated submissions can cause duplicate cost, multiple results, or incorrect write-back. Any automatic retry needs a visible state and a bounded attempt count.

Microphone, screen recording, and file permissions

麦克风、屏幕录制与文件权限

Desktop systems govern microphone, screen recording, folders, and notifications separately. Without permission, no prompt may appear or recording may be silent.

- macOS: open System Settings > Privacy & Security and inspect Microphone and Screen Recording for ZenOffice. Quit and relaunch after a change.
- If ZenOffice is absent, start one recording action to trigger the request. If still absent, place the app in Applications and do not run it from a temporary mounted volume.
- Windows: allow desktop apps under Settings > Privacy & security > Microphone. System loopback also depends on the audio device and driver.
- If open/save fails, check folder permission, free space, read-only state, cloud-sync locks, and whether another application holds the file.

提示 • Tip Use a minimal permission test: blank presentation, ten-second recording, local Desktop save, then add system audio, media, and complex files one variable at a time.

Common shortcuts and recoverable actions

常用快捷键与可恢复操作

Use Command on macOS and Ctrl on Windows/Linux. Save before editing complex objects and test Undo immediately after an AI write-back.

Shortcut	Action
Cmd/Ctrl+N	New
Cmd/Ctrl+O	Open local file
Cmd/Ctrl+S	Save
Cmd/Ctrl+Shift+S	Save As
Cmd/Ctrl+Z	Undo
Cmd/Ctrl+Shift+Z or Ctrl+Y	Redo
Cmd/Ctrl+C / V	Copy / Paste, including supported image paste
Cmd/Ctrl+W	Close current tab/window

- PowerPoint also provides common shortcuts for copy/paste formatting, group/ungroup, duplicate, and zoom; follow the menu display for the current platform.
- Markdown block move, duplicate, and delete shortcuts suit structured editing, but test them on a copy first.
- Undo covers the active editing session; it is not a substitute for versions after close, crash, or external modification.

提示 · Tip Before batch AI edits, page deletion, media replacement, or large format changes, save a clearly named version.

Save, export, and verify in another app

保存、导出与跨应用复核

Correct appearance inside the editor is only the first step. Open DOCX, XLSX, PPTX, PDF, and Markdown in another compatible application to confirm changes were truly written.

1. Save, close, and reopen. Confirm that image edits, equations, tables, citations, and watermarks are not merely temporary preview layers.
2. Verify DOCX in Word/LibreOffice, XLSX in Excel/LibreOffice, PPTX in PowerPoint/LibreOffice, PDF in Preview or Acrobat, and Markdown in a plain-text editor.
3. For PDF export, compare paper size, orientation, margins, full-page position, image width/height, borders, and floating z-order across representative pages.
4. Before delivery, check file size, page count, corruption, passwords, external links, media, fonts, and privacy metadata; retain the source file.

提示 • Tip A minimum acceptance sample includes the first page, an image page, an equation/table page, and the final page. Inspect every page for complex documents.

Privacy, security, and updates

隐私、安全与更新

ZenOffice opens, edits, and saves files locally. ZenMux, web search, and external links create network access. The user decides what content is permitted to leave the device.

- The API Key uses operating-system secure storage and is masked in the UI. Keep it out of screenshots, logs, documents, and repositories; rotate it and limit quota/permissions.
- AI attachments, selections, PDF captures, and prompts may contain sensitive data. Redact first and follow institutional, ethics, contract, and data-transfer rules.
- Download updates only from project Releases. Verify version, filename, and platform; check macOS signing/notarization and scrutinize the source of unsigned Windows builds.
- Do not paste secrets or personal data into a public issue. Follow SECURITY.md and provide the smallest sanitized reproduction.

注意 · Notice Local-first does not mean every AI datum remains on-device. Content relevant to a cloud-model or web-tool request leaves the device.

First-use and delivery checklists

首次使用与交付检查单

Close with two compact checklists: first-use setup focuses on connection and permissions; delivery focuses on editability, compatibility, provenance, and recovery.

First use	Before delivery
☐ Install the right official platform build ☐ Enter the ZenMux key in welcome/settings ☐ Select working text/image models ☐ Test with a short non-private question ☐ Test Save and Save As ☐ Test microphone/screen permissions	☐ Save, close, and reopen ☐ Verify in another compatible app ☐ Check equations, tables, images, citations ☐ Verify PDF pages, ratio, watermark, password ☐ Test media and fonts on target hardware ☐ Keep source, BIB, and version backup

- If AI fails: check key, model, network, proxy, quota, attachment count, and timeout state.
- If a file differs elsewhere: check fonts, page size, object bounds, image ratio, and compatibility of the other app.
- If unsure where to start: select a small passage and test one precise action through the complete loop.

提示 · Tip Official page: <https://bennix.github.io/GenZenMuxAI0ffice/> Source and issues: <https://github.com/bennix/GenZenMuxAI0ffice>

One-minute troubleshooting table

一分钟故障排查表

Match the symptom to its most likely cause and try one minimal action. If it persists, record version, platform, time, model, and a sanitized screenshot.

Symptom	Likely cause	First action
AI timeout / stream stops	Unstable proxy, long task, busy model	Test a short prompt; split work into outline → sections → merge and avoid repeated submissions.
Recognized math does not render	Noisy capture, escaped backslash, broken delimiter	Crop to the formula; check $\dots$, $\$ \dots \$$, and paired environments.
Literature search returns nothing	Query too broad, source unavailable, wrong identifier	Search only DOI, PMID, or arXiv ID; switch OpenAlex/Crossref/PubMed and check connectivity.
Chart or EDA looks wrong	Misaligned selection, unclear type/unit, unexplained outlier	Reselect a small range with headers, define row meaning, units, missing codes, and request evidence rows.
PDF signature becomes invalid	Any page-object change may invalidate a digital signature	Save a copy first. Confirm the approved process before touching a signed contract or official original.
Another app still shows old content	Preview-only change, save failure, or wrong copy	Save, close, reopen, compare full path and modification time, then verify in another compatible app.
Recording lacks microphone/system audio	Permission, relaunch, or platform-source limit	Grant privacy permissions, fully relaunch, and build up from a ten-second blank-deck recording.

提示 · Tip If only one file fails, reproduce with a blank file first. If every file fails, inspect version, permissions, network, and installer source.

MCP: connect the desktop app

MCP: 连接桌面应用

Release supplement · 2026-09-16. MCP exposes application services to an external AI client. These instructions describe current source functionality, not a claim about the published installer.

Prepare and launch

1. Install Node.js 22.12 or newer. Run `npm ci` and `npm run build:all` in the repository root.
2. Generate a local token with `openssl rand -hex 32`. Quit any running ZenOffice instance, then launch from the same terminal:

```
export ZENOFFICE_MCP_PORT=19385
export ZENOFFICE_MCP_TOKEN="YOUR_LOCAL_TOKEN"
./node_modules/.bin/electron apps/shell
```

Add this configuration to the client. Replace both paths with your repository path and use the same generated token:

```
{
  "mcpServers": {
    "zenoffice": {
      "command": "node",
      "args": [
        "--import",
        "/path/GenOffice/node_modules/tsx/dist/loader.mjs",
        "/path/GenOffice/packages/mcp-server/src/cli.ts"
      ],
      "env": {
        "ZENOFFICE_MCP_PORT": "19385",
        "ZENOFFICE_MCP_TOKEN": "YOUR_LOCAL_TOKEN"
      }
    }
  }
}
```

提示 · Tip Reconnect and list tools: expect 38 with the desktop bridge, or 6 independent visualization tools without env. Check `application_status`, then fetch live tabs with `application_list_tabs`. The bridge is disabled by default and listens only on loopback. Tokens must be at least 32 characters; never commit them.

MCP: choose a chart and insert into PDF

MCP: 选图与插入 PDF

The workflow shares chart bindings, AI validation, and rendering with the Excel guide. Use actual data fields and review the preview before insertion.

1. Call `visualization_suggest` with table (columns and rows), `selectedColumns` (allowed field names), and instruction (analysis goal). For Region, Revenue, Notes, select only Region and Revenue to compare revenue by region.
2. The result contains suggestion (chart, axes, fields, title, reason), request (render-ready input), `svg`, and `saved=false`. Only selected-column profiles and the first 30 rows reach the model; validation uses all selected data. Remove sensitive fields before calling if the local client must not receive them either.
3. Review or change `x`, `y`, and title in request, then call `visualization_render_svg`. A file-capable client saves the returned `svg` text as `/absolute/path/chart.svg`; the renderer does not write files.
4. Open a PDF, call `images_targets`, and choose the latest `kind=pdf` target ID. Call `images_insert` with `targetId` and the absolute SVG path in `path`.
5. `inserted=true` confirms insertion on the current PDF page. Move or resize the image, then save in the PDF UI. `saved=false` means the tool did not explicitly save.

Local files and other targets

- PDF Edit → Insert Image also accepts `.svg` files; click the page to place the image. SVG is rasterized; editable vector paths are not retained.
- PNG/JPEG work in Word, PPT, Markdown, and PDF. SVG through MCP currently requires a PDF target. The file limit is 15 MB.
- Untitled Markdown accepts embedded images; saved Markdown uses the adjacent assets directory. Insertion does not imply saving.
- Refresh `images_targets` after opening or closing tabs; Excel refreshes its sharing list automatically. Inspect the target before retrying a timeout.

提示 · Tip AI recommendations use the configured desktop model and may incur provider charges. Use a 1200000 ms client timeout. Without AI, use `visualization_profile` for local recommendations and `visualization_catalog` for bindings. Currently 81 of 104 chart types render; consult availability flags.

MCP: services and document writes

MCP: 服务清单与写回

MCP covers application services, beyond charts. Current coverage is listed below; unlisted editing operations must not be assumed available.

- Visualization (6): catalog, profile, SVG rendering, dashboard SVG, dashboard JSON export and import. Dashboards support 1-6 cards, 1-3 columns, and a shared categorical filter.
- Application (5): status, tabs, activation, open file, create document.
- Projects/chats (8) and knowledge (6): project creation and renaming, file/chat reads, knowledge search and settings.
- Images (2): images_targets, images_insert.
- AI (5): ai_status, ai_chat, visualization_suggest, ai_review, ai_screenwriting.
- Markdown (3): markdown_read, markdown_replace, markdown_save.
- Word (3): word_read_text, word_insert_text, word_save.

Document write sequence

1. Get id from the live tab list. Read Markdown first; pass the exact returned text as expectedText to replace, together with the new text. On conflict, read again and merge. Then save.
2. Read Word text and keep revision. Insert text with id, text, expectedRevision, and position (start/end). This adds literal paragraphs and supports undo before saving. Then call save.
3. Both save tools require an existing file path. Save untitled documents in the UI first. Word refuses to overwrite externally changed files.

提示 • Tip Dedicated screenwriting UI belongs to Word and Markdown, not Excel. Review and screenwriting tools are described next. Official-document formatting, ArtFlow generation, PDF saving, and full Excel/PPT editing remain incomplete in MCP. See docs/mcp-usage.md for full parameters and errors; use the client's live tool list as the source of truth.

AI review and screenwriting: shared workflow

AI 审稿与编剧：统一流程

Word, Markdown, and MCP share model settings, screenwriting skills, and the review committee runner. Confirm the source and task, review the result, then write back.

In the interface

- AI review: choose a profile, report language, and literature search. Inspect independent reviews, chair synthesis, and failures. Reports do not automatically replace the draft.
- Screenwriting belongs to Word and Markdown. Confirm editable source material; choose premise, characters, outline, scene, dialogue, series, adaptation, or diagnosis. Edit the proposal, then append it. If the source document changed, reopen the studio before insertion.

MCP calls

- `ai_review`: `targetId`, `profileId` (default science), `language` (default zh), `literature` (default true). Returns `source`, `members`, `chair`, `literatureEvidence`, `partial`, `inserted=false`, `saved=false`.
- `ai_screenwriting`: `targetId`, `task` (for example `sw-premise-theme`), `medium`, `instruction`. Returns `source`, `content`, `task`, `inserted=false`, `saved=false`. Consult the tool schema for exact task, medium, and review-profile values.
- Only Word and Markdown targets are accepted. MCP reads the whole current unsaved body and rejects documents above 120000 characters. Word supplies available objects/images; Markdown includes up to five readable images. Missing visual or literature evidence is disclosed.

Write back after review

`source` contains the `target`, `path`, `dirty` state, and `source` version. Pass `source.revision` as `expectedRevision` to `word_insert_text`; for Markdown, merge the original text and pass `source.expectedText` to `markdown_replace`. Read and merge again if the source changed.

提示 • Tip Screenwriting loads a pinned screenwriting-skills revision and requires network access on first use. Model calls may incur charges; literature search sends search terms. Cancellation propagates to model and literature requests. Neither MCP service automatically inserts or saves its output.

Start with a real file.



Select context, ask precisely, review the result, then write back to editable objects.

